

ACTIVITY#2(Laboratory)

CODE:

```
1 from Positionallist import Positionallist
2 from LinkedStack import LinkedStack
3
4 usage new *
5 def evaluate_postfix(expression):
6     stack = []
7     for token in expression.split():
8         if token.isdigit():
9             stack.append(int(token))
10        else:
11            b = stack.pop()
12            a = stack.pop()
13            if token == '+':
14                stack.append(a + b)
15            elif token == '-':
16                stack.append(a - b)
17            elif token == '*':
18                stack.append(a * b)
19            elif token == '/':
20                stack.append(a // b)
21            else:
22                raise ValueError(f"Unknown operator: {token}")
23    return stack[0]
24
25 2 usages new *
26 def insertion_sort(arr, ascending=True):
27     for i in range(1, len(arr)):
28         key = arr[i]
29         j = i - 1
30         while j >= 0 and ((arr[j] > key) if ascending else (arr[j] < key)):
31             arr[j + 1] = arr[j]
32             j -= 1
33         arr[j + 1] = key
34     return arr
35
36 if __name__ == "__main__":
37     print("Enter a postfix expression (e.g., '5 2 + 8 3 - * 4 /'):")
38     postfix_expr = input("> ").strip()
```

```
37 try:
38     result = evaluate_postfix(postfix_expr)
39     print("Result of postfix evaluation:", result)
40 except ValueError as e:
41     print("Error:", e)
42
43 print("\nEnter numbers separated by spaces for sorting (e.g., '5 2 9 1 3'):")
44 user_input = input("> ").strip()
45 try:
46     numbers = list(map(int, user_input.split()))
47     sorted_asc = insertion_sort(numbers.copy(), ascending=True)
48     sorted_desc = insertion_sort(numbers.copy(), ascending=False)
49     print("Sorted in Ascending Order:", sorted_asc)
50     print("Sorted in Descending Order:", sorted_desc)
51 except ValueError:
52     print("Error: Please enter valid numbers.")
```

OUTPUT:

```
Enter a postfix expression (e.g., '5 2 + 8 3 - * 4 /'):  
> 5 2 + 8 3 - * 4 /  
Result of postfix evaluation: 8  
  
Enter numbers separated by spaces for sorting (e.g., '5 2 9 1 3'):  
> 1 72 81 25 65 91 11  
Sorted in Ascending Order: [1, 11, 25, 65, 72, 81, 91]  
Sorted in Descending Order: [91, 81, 72, 65, 25, 11, 1]  
  
Process finished with exit code 0
```